

Tribhuvan University, Institute of Science and Technology



Comparing CatBoost, TabPFN, and TabFM for Explainable Hair Fall Risk Prediction

Dissertation Proposal

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Project at a glance

Problem	Hair fall risk is hard to judge, and most prediction models cannot explain their answers.
Objectives	1. Benchmark CatBoost, TabPFN, and TabFM. 2. Explain every model with SHAP.
Research gap	Foundation models were never tested on hair fall data; CatBoost was never tuned; no side-by-side explanations.
Data	21,606 records, 20 features (blood and body measurements, lifestyle, clinical), 3 risk tiers: Low, Moderate, High.
Method	Same split for all models; CatBoost tuned, TabPFN and TabFM without tuning; SHAP; accuracy, macro-F1, ROC-AUC, McNemar's test.
Expected outcome	A fair comparison, a strong tuned baseline, TabFM limits recorded, and explanations checked against known biology.
Time	4 months: proposal work (Months 1 and 2) and final work (Months 3 and 4).

Problem statement

- Hair loss risk is hard to judge: hormones, nutrition, stress, illness, and habits act together.
- Assessment often relies on self-reporting or late clinical checks, which are subjective and hard to reach in rural areas.
- Earlier prediction studies used classical algorithms and report very different accuracies (about 50% to 100%).
- Newer tabular foundation models (TabPFN, TabFM) have not been tested on hair fall data.
- Both gradient boosting and foundation models are hard to explain.

Objectives

1. To implement and benchmark CatBoost, TabPFN, and TabFM for predicting multi-tier hair fall risk from structured clinical and lifestyle data.
2. To provide transparent, feature-level decision-support insights for every model through SHAP-based explainability, so that each predicted risk tier can be traced to specific indicators.

Background: what causes hair loss

Factor	What it does	Source
Hormones and heredity	Inherited sensitivity to androgens; family history is a key indicator	Agaoglu et al. (2021)
Nutrition and iron	Shortage of iron, zinc, and vitamins weakens follicles	Guo & Katta (2017); Lin et al. (2023)
Thyroid function	Thyroid hormones regulate the hair cycle	Hussein et al. (2023)
Psychological stress	Pushes follicles into the resting phase early	Bai et al. (2026)
Illness and infection	Can trigger heavy shedding (telogen effluvium)	Cline et al. (2021)

Trend: hair loss is reported more often and at a younger age, and it affects self-confidence and quality of life.

Literature: what others did

Study	Data	Models	Main result
Khatun et al. (2022)	Survey, 610 people	SVM, KNN, LR, RF, XGBoost	XGBoost 92.62%
Sai et al. (2023)	Hair fall data	SVM, KNN, DT, RF, LR, ensemble	Ensemble best
Kumar et al. (2025)	2,000 records	RF, XGBoost, CatBoost, LightGBM	RF 100%, CatBoost 49.5%
Siامي & Azis (2025)	Multi-factor data	LR, DT, RF, GB, XGBoost, voting	About 50% at best
Leema et al. (2025)	Survey, 750 people	LSTM, RF, TFT, ARIMAX	TFT 97.5%
Shakeel et al. (2021); Sayyad et al. (2022)	Hair images	SVM, KNN, VGG with SVM	91.4% and 98.31%

LR = Logistic Regression, DT = Decision Tree, RF = Random Forest, GB = Gradient Boosting

Literature: the three model families

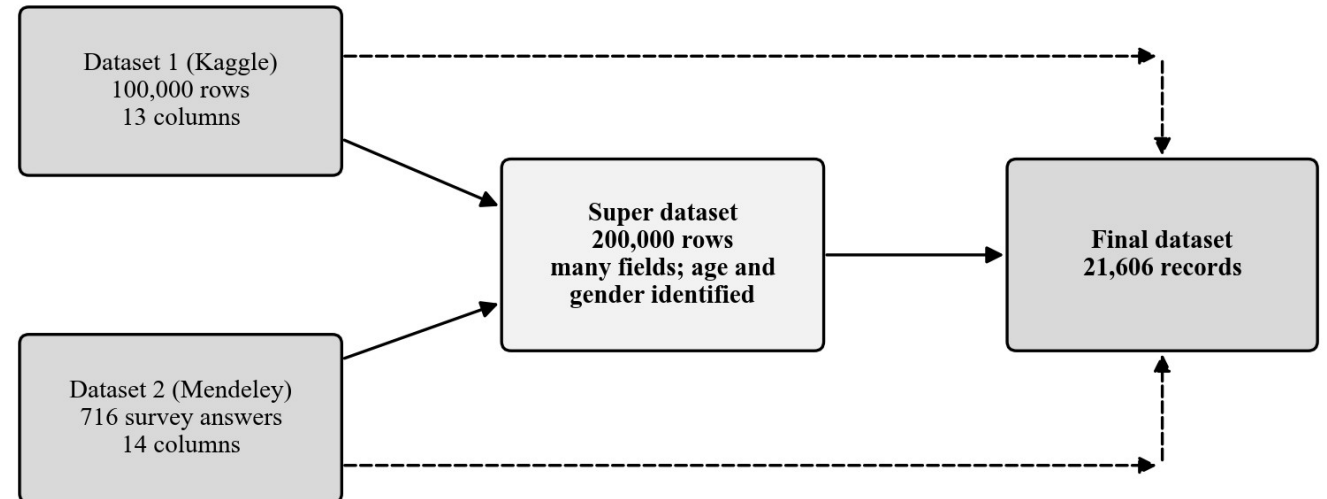
- CatBoost (Prokhorenkova et al., 2018): gradient boosted trees with ordered encoding of categorical answers.
- TabPFN (Hollmann et al., 2025): transformer pretrained on synthetic tables; predicts in one forward pass; best on small tables.
- TabFM (Kong et al., 2026): 400-million-parameter foundation model; first among default foundation models on 51 TabArena datasets.
- An independent check of TabFM reported software defects and memory limits on larger tables (Pandey, 2026).
- SHAP (Lundberg & Lee, 2017): explains any model by the contribution of each feature.

Research gap

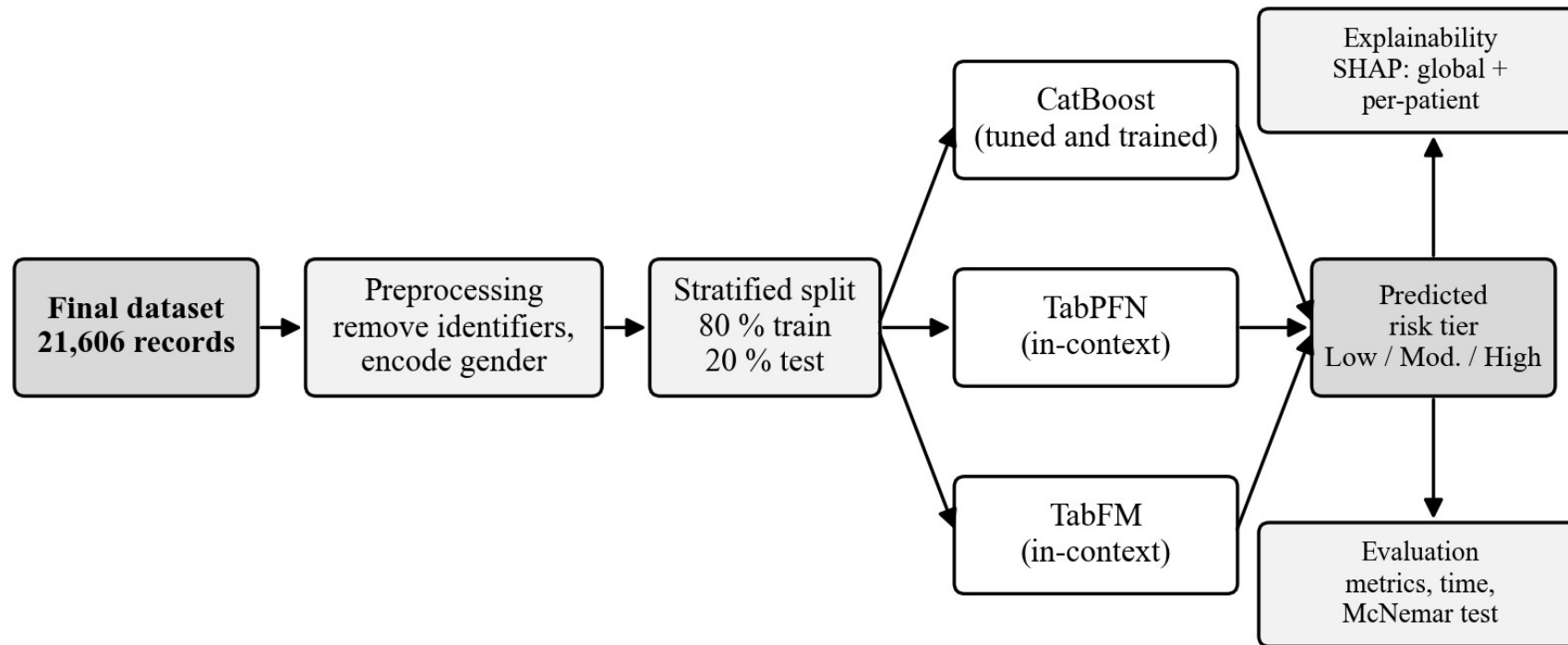
- Hair loss prediction studies used only classical algorithms, and accuracies differ widely.
- CatBoost was compared only once, and it was not tuned.
- TabPFN and TabFM have not been tested on hair loss data.
- TabFM has only one independent check, which reported defects and memory limits.
- No study explains a boosted-tree model and foundation models side by side.

Data

- Dataset 1 (Kaggle): 100,000 records, 13 columns of measurements and hair fall (0 to 5).
- Dataset 2 (Mendeley): survey of 716 people, 14 columns.
- Super dataset: 200,000 records with all fields of both and extra fields.
- Final dataset: 21,606 records, 20 features, target Low 45%, Moderate 35%, High 20%.



Research framework



Same split and seed for all models. Training sizes: 500, 2,000, and all 17,284 records.

Models

Model	How it works	Training and tuning
CatBoost	Many small symmetric decision trees, each correcting the previous ones	Trained and tuned: grid search with 5-fold cross-validation
TabPFN	Transformer reads training rows and the patient, predicts in one pass	None; weights are fixed
TabFM	Transformer with column and row attention, about 400 million parameters	None; weights are fixed

TabPFN and TabFM will run on a Kaggle GPU; CatBoost and the statistics on a Mac.

Explainability and evaluation

SHAP for all three models:

TreeExplainer for CatBoost

KernelExplainer for TabPFN and TabFM

Global ranking and per-patient explanations

Evaluation on the same test set:

Accuracy, macro-precision, macro-recall,
macro-F1

ROC-AUC (one-versus-rest)

Confusion matrix

McNemar's test and Wilson 95% intervals

Expected outcomes

- A fair comparison of three model families on the same hair fall data.
- A properly tuned CatBoost as a strong baseline.
- TabPFN and TabFM competitive without tuning, especially with small training sets (to be tested).
- The practical limits of TabFM recorded on a real health dataset.
- SHAP explanations that agree with known causes: iron, stress, family history.

Schedule: 4 months

